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ABSTRACT

Thermoelectric cooling using Peltier modules presents a compact, energy-efficient alternative to conventional vapor-compression refrigeration. This project investigates the performance of a TEC12706 Peltier module integrated with an Arduino-based control system, W1209 temperature controller, DS18B20 digital temperature sensor, and a forced-air heat dissipation system. The system uses a 12V–5A DC power supply to drive the module and associated electronics.

The objective of the project is to analyze heat transfer behavior, cooling efficiency, and temperature control accuracy in a low-cost thermoelectric cooling setup. The TEC module creates a temperature differential across its surfaces when powered, resulting in a cold and hot junction. A PWM cooling fan paired with an H410R heat sink enhances heat rejection to maintain the temperature gradient. Performance was evaluated through experimental testing, measuring cooling rate, stable temperature, and power consumption.

The results indicate that with proper heat dissipation, the TEC12706 can achieve significant cooling below ambient temperature within a short period. The system demonstrates potential applications in portable coolers, electronics cooling, small refrigeration units, and laboratory thermal control systems. This project integrates principles of heat transfer, thermoelectric physics, electronics control, and mechanical assembly to present an efficient and scalable cooling solution.

CHAPTER 1

INTRODUCTION

Refrigeration technologies have evolved significantly, yet conventional systems still rely on compressor-based cycles involving refrigerants, high-pressure components, and bulky mechanical assemblies. Thermoelectric cooling, based on the Peltier effect, offers an alternative through solid-state heat pumping without moving parts. Peltier modules are compact, lightweight, environmentally friendly, and suitable for small-scale, precise cooling applications.

The TEC12706 thermoelectric module used in this project operates when DC current passes through semiconductor junctions, causing one side to absorb heat and the opposite side to reject heat. This creates a temperature difference that can be used for cooling or heating. However, effective thermal management is essential, since heat accumulates on the hot side and can reduce efficiency if not dissipated properly.

In this project, a complete thermoelectric cooling system was developed using a TEC12706 module, an H410R heatsink, PWM cooling fan, Arduino Uno R3, W1209 temperature controller, and DS18B20 digital temperature sensor. The system enables automatic temperature monitoring and control while maintaining consistent thermal performance.

The purpose of this study is to analyze the refrigeration capability, temperature control accuracy, and practical applicability of a Peltier-based cooling system in real-world scenarios.

CHAPTER 2

PROBLEM STATEMENT

Traditional refrigeration systems are efficient but rely on large mechanical components, refrigerant gases, and high power consumption. These systems are unsuitable for small-scale applications such as portable cooling units, electronics thermal management, or localized spot cooling. Additionally, conventional refrigeration produces vibrations, noise, and environmental concerns due to refrigerant usage.

The challenge addressed in this project is:

“To design a compact, low-power, environmentally friendly thermoelectric cooling system using a Peltier module and analyze its performance, heat dissipation, and temperature control accuracy.”

Key engineering issues include:

- Managing heat accumulation at the hot side of the Peltier module
- Ensuring stable voltage/current supply
- Controlling temperature precisely
- Achieving efficient thermal conduction between components
- Evaluating cooling capacity and steady-state temperatureThe project aims to create a functional prototype demonstrating these concepts using widely available low-cost components.

CHAPTER 3

LITERATURE SURVEY

Several studies over the past decade have analyzed thermoelectric cooling technology due to its simplicity and ecological benefits. Researchers have focused on improving heat dissipation, module efficiency, and control strategies.

• M. Rowe (2008) analyzed thermoelectric device performance and highlighted that maintaining a low hot-side temperature is crucial for maximizing cooling efficiency.

• R. Chen et al. (2014) demonstrated that forced convection cooling on the hot side significantly improves the coefficient of performance of TEC modules, supporting the use of fans and heatsinks.

• N. Babu & M. Priya (2017) evaluated the TEC12706 module and showed that cooling efficiency depends strongly on heat sink design and thermal contact resistance.

• Studies have also explored the integration of microcontrollers, confirming that automated temperature monitoring increases stability and reduces module degradation over time.

The literature suggests that while thermoelectric cooling is less efficient than compressor-based systems, it excels in small-scale, precise, vibration-free, and environmentally safe applications—aligning directly with the goals of this project.

CHAPTER 4

Existing System

Conventional cooling and refrigeration systems predominantly rely on the vapour compression refrigeration cycle, which has remained the industry standard for decades due to its relatively high efficiency and ability to achieve large temperature differentials. These systems consist of a compressor, condenser, expansion valve, and evaporator working together to circulate a refrigerant. While extremely effective for large-scale cooling, these systems bring several drawbacks when considered for small or portable applications. They require mechanical moving parts, regular maintenance, refrigerant refilling, leak-proof enclosures, and are associated with noise and vibrations. The presence of refrigerants such as R134a or R600a also poses environmental risks in case of leakage, contributing to global warming due to their high Global Warming Potential (GWP).

In the electronics and compact device markets, conventional refrigeration becomes impractical due to its bulk, weight, and complexity. For example, cooling microprocessors, battery packs, medical storage capsules, and laboratory samples require precision cooling with minimal physical footprint. In such cases, heat sinks and fans are commonly used; however, their performance is limited to keeping temperatures close to or slightly below ambient. They cannot achieve sub-ambient cooling essential for specialized applications.

Another existing method is liquid cooling, which uses a pump to circulate coolant through a heat exchanger and radiator. While effective for high-performance computers and industrial electronics, liquid cooling systems are costly, require leak-proof assembly, and demand periodic maintenance. Their dependency on pumps and fluid flow also introduces the risk of mechanical failure. For small-scale refrigeration, such complexity is neither economically viable nor energy-efficient.

Additionally, thermosiphon cooling and phase-change materials (PCM) have been used in some portable devices. Although these systems can temporarily absorb heat, their behaviour is passive and limited to short-duration temperature stabilization. They do not offer active or continuous cooling.

Therefore, there exists a gap in cooling technology for small, silent, portable, low-power systems that require sub-ambient temperature performance. Thermoelectric cooling using the Peltier effect fills this gap by offering solid-state heat pumping with no moving parts. Yet, traditional thermoelectric systems also face limitations: poor heat dissipation, inadequate thermal management, and lack of automatic control reduce their efficiency.

Hence, the existing system, whether vapour-compression, air cooling, or liquid cooling, does not fully satisfy the emerging needs for miniaturized, energy-efficient, and ecologically sustainable cooling. For academic, laboratory, and compact commercial applications, an improved approach

CHAPTER 5

PROPOSED SOLUTION I — BASIC PELTIER COOLING SYSTEM

The first proposed solution presents a fundamental thermoelectric cooling setup designed around the TEC12706 module operating on the Peltier principle. This solution aims to develop a simple and functional cooling device without the inclusion of automatic temperature feedback or complex control circuitry. The purpose of this stage is to understand the fundamental heat transfer behaviour of the Peltier module and evaluate how effectively it can generate cooling when powered directly under fixed electrical conditions.

In this configuration, a 12V 5A DC power supply powers the TEC12706 module. When current flows through the semiconductor junctions inside the TEC, one surface absorbs heat (cold side) while the opposite surface rejects heat (hot side). However, without proper heat dissipation, the temperature difference deteriorates rapidly. To address this, the hot side is firmly attached using thermal paste to a H410R Cooling Master aluminum heatsink, which provides a large surface area for conductive and convective heat removal. Attached to this heatsink is a PWM-controlled cooling fan, which facilitates forced air convection, enhancing the transfer of heat to the surroundings.

The cold side, meanwhile, is left exposed or connected to a small chamber or metal block for cooling demonstration. The goal is to observe the drop in temperature and the time required to achieve a measurable cooling effect. This basic system allows analysis of parameters such as thermal contact resistance, heat sink performance, and cooling rate.

One significant limitation of this simple setup is the absence of temperature regulation. The TEC module continues to draw high current and operate at full capacity until manually turned off. Over time, the module may overheat, especially if the heatsink is insufficient or airflow is obstructed. This reduces the cooling efficiency and increases the risk of module damage. Furthermore, without feedback control, temperature fluctuations are common, making the system unsuitable for applications requiring stability.

Despite its limitations, Proposed Solution I is valuable for understanding the foundational behaviour of the TEC12706 under different loads and environmental conditions. It establishes a baseline performance dataset, highlighting the need for improved heat management and automated control to utilize thermoelectric cooling effectively. The observations gained from this setup form the basis for the development of a more sophisticated system presented in Proposed Solution II.

CHAPTER 6

PROPOSED SOLUTION II — AUTOMATED TEMPERATURE-CONTROLLED SYSTEM

Building on the insights gained through the basic cooling system, Proposed Solution II introduces a more advanced and practically viable thermoelectric cooling system featuring automated temperature monitoring and control. The primary objective of this solution is to enhance cooling efficiency, prevent overheating of the Peltier module, and maintain stable temperatures for prolonged applications.

This solution integrates several key components: an Arduino Uno R3 microcontroller for intelligent decision-making, a DS18B20 digital temperature sensor for accurate thermal measurement, a W1209 temperature control module for relay-based switching, and a 4.7kΩ resistor used as a pull-up resistor for one-wire communication with the sensor.

The Arduino receives real-time temperature readings from the DS18B20 sensor placed near the cold plate or inside the cooling chamber. Based on the programmed threshold, the Arduino sends control signals to the W1209 module, which activates or deactivates the TEC12706 by switching the relay. This ensures the module operates only when necessary, preventing unnecessary power usage and prolonging the module's lifespan.

Meanwhile, the PWM fan and heatsink arrangement from Solution I is retained to manage the heat load on the hot side. However, in this enhanced setup, the temperature controller ensures that the fan can also be modulated depending on the heat generation. This improves efficiency by not running the fan at full speed continuously, reducing noise and power consumption.

One of the biggest advantages of this solution is its precision. The DS18B20 sensor provides ±0.5°C accuracy, enabling the system to maintain a setpoint temperature within a narrow range. This makes the system ideal for applications requiring stability, such as miniature refrigerators, electronic component cooling, biological sample preservation, and small laboratory devices.

Additionally, this solution allows for data logging and expansion. Temperature curves, power usage patterns, and thermal response times can all be recorded and analyzed. The microcontroller can be further extended to include LCD displays, IoT connectivity, or adaptive control strategies such as PID tuning.

Overall, Proposed Solution II offers a robust, intelligent, safe, and energy-efficient thermoelectric cooling system that overcomes the limitations identified in the basic system.

CHAPTER 7

RECOMMENDED SOLUTION

After evaluating both proposed systems, the second solution—the automated temperature-controlled thermoelectric cooling system—emerges as the recommended choice due to its superior performance, safety, energy efficiency, and usability in real-world environments. While the first solution serves as an important baseline for understanding fundamental thermoelectric behaviour, it cannot meet the reliability or precision required for practical cooling applications.

Solution II introduces automation, feedback control, and system protection features that are crucial for stable thermoelectric operation. By integrating Arduino Uno R3 with the DS18B20 sensor and W1209 controller, the system achieves consistent temperature monitoring and intelligent control, preventing overheating and ensuring the Peltier module functions within optimal operating parameters. This reduces energy wastage and prolongs the life of the module, addressing one of the key limitations of thermoelectric technology.

The recommended solution also allows scalability and customization. For example, temperature setpoints can be programmed based on application needs, whether the cooling target is an electronic component, a biological sample, a beverage container, or a temperature-sensitive device. Additionally, the integration of a PWM fan with adjustable speeds ensures efficient heat dissipation while keeping power consumption low.

From an engineering perspective, this system better demonstrates applied concepts of heat transfer, thermal conduction, convection enhancement, microcontroller programming, electronic control, and system integration. It provides a holistic representation of multidisciplinary engineering practice, making it an ideal solution both academically and practically.

Economically, the recommended solution achieves higher energy savings by running the TEC module only when required. This is particularly beneficial in battery-operated or portable devices where power efficiency is critical. Environmentally, the absence of harmful refrigerants and the overall low carbon footprint of the system make it an attractive eco-friendly alternative to conventional cooling technologies.

For these reasons, Proposed Solution II is strongly recommended as the optimal system design.

CHAPTER 8

ARCHITECTURE OF PROPOSED SYSTEM

The architecture of the automated thermoelectric cooling system integrates thermal, electrical, and control subsystems into a cohesive functional model. At the core of the architecture is the TEC12706 thermoelectric module, which acts as the primary cooling mechanism. This module is mounted between two substrates: the cold plate and the hot-side heatsink. Thermal paste is used on both surfaces to minimize contact resistance and enhance heat transfer.

The hot-side subsystem consists of the H410R aluminum heatsink fitted with a PWM cooling fan. This subsystem ensures that heat rejected from the TEC is removed efficiently to maintain a high temperature gradient across the module. The performance of the entire system is heavily dependent on maintaining effective heat dissipation, as thermal saturation would significantly reduce the cooling output.

The cold-side subsystem includes the cooling plate or chamber that receives the cooling effect. The DS18B20 temperature sensor is positioned strategically on this side to capture accurate temperature readings. This allows the controller to make precise decisions about activating or deactivating the Peltier module.

The control subsystem includes the Arduino Uno R3 microcontroller, which acts as the brain of the system. It receives input signals from the DS18B20 sensor through a one-wire communication method stabilized using a 4.7kΩ pull-up resistor. The Arduino processes temperature data according to programmed logic and sends output signals to the W1209 temperature controller.

The W1209 module, equipped with an onboard relay, switches the power to the TEC module based on the Arduino's instructions. This relay-based switching isolates the high-current TEC supply from the low-power control circuitry, ensuring safe and reliable operation.

Finally, the power subsystem consists of the 12V 5A DC supply, which provides sufficient current to power the TEC module, cooling fan, and control electronics. Power distribution is carefully managed to prevent voltage drops or uneven supply.

This architecture ensures seamless operation by integrating sensing, decision-making, and actuation in a closed-loop control system. The modular layout also enables easy expansion for future enhancements such as PID control, display interfaces, or IoT connectivity.

CHAPTER 9

TECHNICAL REQUIREMENTS

The technical requirements of the thermoelectric cooling project encompass all electrical, mechanical, and thermal components necessary to assemble, operate, and evaluate the performance of the system. Each component plays a critical role in achieving efficient cooling, stable control, and reliable system behaviour.

Power Supply: A 12V, 5A regulated DC power supply is essential. The TEC12706 module draws significant current (up to 6A peak), and inadequate supply leads to voltage drops, reduced cooling, and overheating. The power supply must include overload protection, short-circuit protection, and stable DC output.

Peltier Module (TEC12706): The module must be genuine and capable of delivering a temperature differential of up to 60°C under controlled conditions. Proper polarity must be ensured during installation, as reversing polarity reverses the cooling and heating sides.

Heatsink and Cooling Fan: The H410R Cooling Master heatsink and PWM fan are required to efficiently dissipate the heat generated on the hot side. The fan must provide adequate airflow, preferably above 2000 RPM, while remaining energy-efficient and compatible with 12V operation.

Thermal Interface Materials: High-quality thermal paste or thermal pads are essential to reduce thermal contact resistance between the TEC module and heatsink/cold plate. Poor contact reduces system performance significantly.

Sensor System: The DS18B20 digital temperature sensor provides accurate measurements essential for feedback control. A 4.7kΩ resistor is used as a pull-up resistor to ensure proper communication in a one-wire data system.

Microcontroller: The Arduino Uno R3 is chosen for its ease of programming, availability of libraries, and compatibility with digital sensors. It also allows future expansion to IoT-based monitoring or PID control.

Control Module: The W1209 module provides relay-based switching for the Peltier module, preventing the microcontroller from handling high current directly. It adds electrical isolation and ensures safe operation.

Electrical Wiring and Connectors: High-quality insulated wires with correct gauge ratings are necessary to prevent overheating. Solder joints must be reliable, and connectors should be tight and secure.

Mechanical Components: Mounting brackets, screws, clamps, acrylic sheets or aluminum plates, and insulation materials are needed to assemble the system mechanically.

CHAPTER 10

APPLICATIONS

Thermoelectric cooling systems based on the Peltier effect have numerous applications across various industries due to their solid-state operation, compact design, and ability to achieve sub-ambient temperatures with high precision. This chapter explores the diverse applications of the cooling system developed in this project.

One of the primary applications is in electronics cooling, particularly for processors, graphic units, and power electronics. Traditional cooling systems can only lower component temperatures close to ambient, but Peltier modules can provide sub-ambient cooling, which is advantageous for overclocking, high-performance computing, and laboratory testing. The precise temperature control offered by the integrated sensor and microcontroller system enhances safety and device longevity.

In portable refrigeration, thermoelectric coolers are widely used for small refrigerators, beverage coolers, and mini-fridges. Their lightweight and vibration-free operation make them ideal for camping, automotive consoles, portable medical boxes, and personal use refrigerators. Unlike vapour-compression systems, they don't require refrigerant gases or mechanical compressors, making them environmentally responsible solutions.

Thermoelectric systems are also heavily used in medical and laboratory equipment, where stable temperature environments are required. Examples include vaccine storage boxes, blood transportation containers, biological sample preservation units, and temperature-controlled incubation chambers. The precise control mechanism and solid-state reliability make TEC-based cooling highly suitable for such sensitive applications.

In industrial settings, thermoelectric modules are used for spot cooling, where only a small, localized area needs to be cooled instead of an entire environment. Examples include cooling laser diodes, sensors, power transistors, and LED modules. Similarly, its application in optical communication systems ensures stable operating temperatures for photodetectors and laser transmitters.

Consumer applications also benefit from thermoelectric technology. Devices like car seat coolers, food warming and cooling boxes, cosmetic coolers, and smart water dispensers employ Peltier modules. The silent operation is a major advantage in household environments where noise reduction is preferred.

Furthermore, Peltier coolers are used in thermal cycling equipment such as PCR machines in molecular biology. Their ability to rapidly change temperature by reversing polarity makes them ideal for processes requiring rapid heating and cooling cycles.

The system developed in this project, with its intelligent temperature control and robust heat dissipation system, is suitable for all the above applications. Its modular design allows customization depending on the end-use scenario. The incorporation of microcontroller-based

CHAPTER 11

FUTURE SCOPE

Thermoelectric cooling technology holds significant potential for future advancements, and the system developed in this project can be expanded in numerous ways to enhance its efficiency, versatility, and practical relevance. This chapter highlights potential areas of improvement and exploration.

One major direction for future development is the integration of multi-stage Peltier modules. These cascaded systems can achieve much lower temperatures than single-stage modules by stacking TECs in series. This would make the system capable of reaching temperatures required for cryogenic storage, advanced laboratory instruments, and deep cooling applications.

Another promising enhancement is the implementation of PID (Proportional–Integral–Derivative) control instead of simple relay-based switching. PID control would enable smoother temperature regulation, minimize overshoot, and enhance stability. This is particularly important for applications requiring precise thermal conditions such as DNA amplification, semiconductor testing, and sensitive biological experiments.

Additionally, future versions of the system could incorporate IoT-based monitoring and control, enabling users to view temperature data remotely through smartphones or computers. This would be beneficial for applications such as vaccine transport, server cooling, or large-scale monitoring of multiple thermoelectric devices in industrial settings.

The efficiency of Peltier modules can also be enhanced by improving the hot-side heat dissipation system. Future work may include the use of liquid cooling systems, heat pipes, or advanced materials such as graphene thermal pads to dramatically improve heat removal. This would significantly improve the cooling performance and energy efficiency of the system.

In terms of power management, incorporating renewable energy sources, such as solar panels, could make the system suitable for remote or rural applications where electricity is unreliable. Portable solar-powered coolers could serve important roles in humanitarian aid, medical outreach programs, and outdoor activities.

On the materials front, advancements in thermoelectric materials such as bismuth telluride alloys, skutterudites, and nanostructured materials promise higher efficiency. Incorporating such materials in future systems could reduce power consumption while increasing cooling capacity.

Finally, scaling the system into a commercial prototype would involve ergonomic design improvements, aesthetic enclosures, user-friendly interfaces, and integration of safety mechanisms such as thermal shutdown and overcurrent protection.Overall, the future scope for thermoelectric cooling technology is vast, and this project forms a solid foundation for further innovation.

CHAPTER 12

CONCLUSION

This project successfully demonstrates the development of a compact thermoelectric cooling system using the TEC12706 module. Experimental results show that with appropriate heat dissipation and automated temperature control, significant cooling below ambient conditions can be achieved. The system is environmentally friendly, vibration-free, and suitable for applications requiring precise, low-power cooling.

The combination of Peltier technology with microcontroller-based control offers a reliable, scalable alternative to conventional refrigeration methods for small-scale applications.

CHAPTER 13

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